

Problem 1. Translate these statements into logic using the provided propositions.

$v \equiv$ The message is scanned for viruses.

$s \equiv$ The message is marked as spam.

$k \equiv$ The message was sent from a known system.

$b \equiv$ The sender is blocked.

- a) The message is scanned for viruses or marked as spam.
- b) A message from a blocked sender goes scanned for viruses.
- c) The sender is blocked and the message is scanned for viruses only if the message was sent from an unknown system.
- d) If the message is sent from an unknown system, either the sender is blocked or the message is marked for spam. In both cases, it is scanned for viruses.

Answer.

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Problem 2. Prove the following by using **logical equivalences** (i.e., using direct manipulation). For each step, state which single logical equivalence was used.

- a) $\neg(p \rightarrow \neg q) \equiv (p \wedge q)$
- b) $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Answer.

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Problem 3. Use the truth tables to:

- a) Show that $p \vee (p \wedge q) \equiv p$
- b) Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent

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Problem 4. Let $M(x) \equiv x \text{ visits the Carnegie Museum}$ and $A(x) \equiv x \text{ is an art student}$. Let the domain of each be “all students at Pitt”.

- a) Translate the following statement into English.

$$\forall x [M(x) \rightarrow A(x)]$$

- b) Translate the following into predicate logic.
“Not all art students have visited the Carnegie Museum”.

- c) Explain why the statement “Some art students visit the Carnegie Museum”
cannot be translated as:

$$\exists x [A(x) \rightarrow M(x)]$$

Answer.

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Problem 5. Explain the following logical paradoxes. That is, explain why each one cannot be assigned a specific value.

- a) “The next statement is true. The previous statement is false.”
- b) “If this sentence is true, then all statements are true.”
- c) Define two propositional variables as $p \equiv q$ is false, and $q \equiv p$ is false.

Answer.